

CSCI212: Lecture #15

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Goals of this lecture

- Using text in GUIs: input & output in GUIs, e.g., typing messages on screen for the user.
- Learning about Layout managers
- Learn a bit more about the class **String** (especially in working with text based operations)
- More practice with exceptions; users will do things that cause problems!

Let's go over the **JTextArea** class (if it begins with "**J**" then it is a **Swing** class).

JTextArea: an area to hold text from the user or from the GUI.

JLabel: the text that the GUI writes.

JScrollPane: can hold an expandable text area.

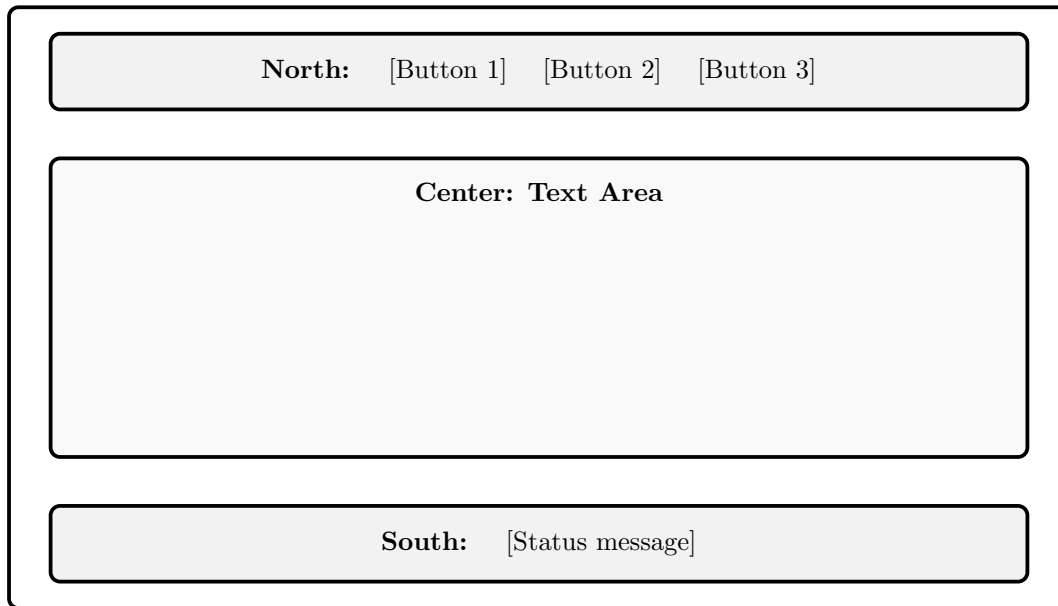
Arranging components on the GUI requires a layout manager. There are two that we have covered in class, the **FlowLayout** (the default manager) and the **BorderLayout** which separates the layout into 5 distinct positions: **North**, **South**, **East**, **West**, and **Center**. However, in each position of the BorderLayout, we can place **at most 1** component.

JPanel: Provides additional organizational flexibility.

Typical GUI Organization

- We make a class for the GUI
- A constructor : lists what goes on the screen, adds listeners with lambdas to respond to the user.
- Supply methods to code responses to the user
- a **main method** will run the GUI.

Goal (our schematic of the GUI)



In this case, the center text area can be implemented using a **JTextArea** in a **JScrollPane**. The GUI can be formatted under a **Border Layout**.

Implementation of the schematic:

```
1 import javax.swing.*;
2 import java.awt.*;
3
4 public class NotesApp {
5
6
7     private JTextArea area;
8     private JLabel status;
9     private JPanel buttonPanel;
10
11
12     public NotesApp() {
13         JFrame frame = new JFrame();
14         frame.setLayout(new BorderLayout());
15
16         area = new JTextArea(10, 30);
17         status = new JLabel("Ready");
18         buttonPanel = new JPanel();
19
20         frame.add(buttonPanel, BorderLayout.NORTH);
21         frame.add(new JScrollPane(area), BorderLayout.CENTER);
22         frame.add(status, BorderLayout.SOUTH);
23
24         JButton wordButton = new JButton("Word Count");
25         JButton charButton = new JButton("Character Count");
26         JButton avgButton = new JButton("Average Length");
27         JButton clearButton = new JButton("Clear");
28
29         buttonPanel.add(wordButton);
30         buttonPanel.add(charButton);
31         buttonPanel.add(avgButton);
32         buttonPanel.add(clearButton);
33     }
34 }
```

```

33     wordButton.addActionListener(e -> handWordCount());
34     charButton.addActionListener(e -> handleCharCount());
35     avgButton.addActionListener(e -> handleAverage());
36     clearButton.addActionListener(e -> handleClear());
37
38
39     styleUI(frame, wordButton, charButton, avgButton, clearButton);
40
41     frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
42     frame.setVisible(true);
43 }
44
45 // Functionality will be included in the following lecture.
46 public void handWordCount() {}
47 public void handleCharCount() {}
48 public void handleAverage() {}
49 public void handleClear() {}
50
51
52 private void styleUI(JFrame frame, JButton wordButton, JButton charButton, JButton
53     avgButton, JButton clearButton) {
54
55     // Frame settings
56     frame.setSize(500, 400);
57     frame.setLocationRelativeTo(null); // center on screen
58
59     // Text area styling
60     area.setFont(new Font("Monospaced", Font.PLAIN, 14));
61     area.setLineWrap(true);
62     area.setWrapStyleWord(true);
63
64     // Status label styling
65     status.setFont(new Font("Arial", Font.BOLD, 12));
66     status.setHorizontalAlignment(SwingConstants.CENTER);
67
68     // Button styling
69     JButton[] buttons = {wordButton, charButton, avgButton, clearButton};
70
71     for (JButton b : buttons) {
72         b.setFocusPainted(false);
73         b.setFont(new Font("Arial", Font.BOLD, 12));
74     }
75
76     // Panel styling
77     buttonPanel.setLayout(new FlowLayout(FlowLayout.CENTER, 10, 10));
78
79     public static void main(String[] args) {
80         SwingUtilities.invokeLater(() -> new NotesApp());
81     }
82 }

```